

EDUCATION

Bunker Hill Community College

A.S. in Computer Science; GPA: 4.0/4.0

Boston, MA

Jun. 2022 – Dec. 2023

University of Massachusetts Lowell

B.S./M.S. in Computer Science; GPA: 4.0/4.0

Lowell, MA

Jan. 2024 – May. 2025

SKILLS

- **Programming Languages:** Java (4 yrs), Javascript/Typescript (4 yrs), Python (3 yrs), Go, C++, Kotlin, PHP, Lua, Rust, Bash, Markdown, SQL, HTML/CSS, LaTeX, XML, TOML
- **Frameworks and Libraries:** Spring Boot, Node.js, React, MUI, NestJS, Vite, JUnit, Nest.js, Drizzle ORM
- **Databases and APIs:** MySQL (3yrs), PostgreSQL, MongoDB, Redis, HTTPS, RESTful, GraphQL, WebSockets, gRPC
- **Tools and Platforms:** Git, GitHub, Nginx, systemd, Docker, AWS EC2, AWS Lightsail, POSIX, npm/pnpm, Neovim, Postman
- **Machine Learning:** Numpy, Matplotlib, Scikit-learn, PyTorch

WORK EXPERIENCE

Juyingtong

Backend Developer

Zhuhai, China

Nov. 2018 – Feb. 2020

- Developed the backend for a WeChat mini-program for conference management using ThinkPHP 5 and MySQL, with Redis-backed session authentication and RESTful APIs for user and event operations.
- Built a location-aware attendee check-in system using QR code scanning, encrypted conference metadata, WeChat geolocation validation, and real-time attendance recording.
- Integrated the WeChat Pay API to support secure, real-time payment processing for conference registration and related services.
- Deployed the application on Alibaba Cloud Linux servers with Nginx reverse proxy and load balancing, using systemd for process management.

Take 2 Health

Backend Developer Intern

Shenzhen, China

Mar. 2021 – Sep. 2021

- Engineered the backend for a blood specimen management platform using Midway.js, implementing JWT-based authentication and RESTful APIs for user and specimen operations.
- Authored Swagger API documentation to streamline integration and testing, and collaborated with frontend engineers to ensure reliable interface behavior.
- Architected a microservice-based system using gRPC for high-performance inter-service communication, improving scalability and modularity.

UMass Lowell - Miner School of Computer and Information Sciences

Software Engineering Intern

Lowell, MA

May. 2025 – Dec. 2025

- Migrated 25k+ AEPW (Alliance to End Plastic Waste) records into a redesigned PostgreSQL schema, cutting query times by about 70% and improving data integrity.
- Designed and implemented AUDA, a modular Python analytics platform for reproducible data migration, preprocessing, forecasting, and visualization workflows.
- Built and evaluated machine learning workflows for sparse country-level time series, including Isolation Forest anomaly detection, Random Forest feature importance, and SVR forecasting with automated hyperparameter optimization.
- Developed a Typer-based CLI to automate end-to-end analytics workflows, improving reproducibility, maintainability, and developer productivity.

PROJECTS

- **Course Sync:** A serverless course synchronization CLI for instructors and students using **Go**, **Cobra**, **Git**, and **JSON**-backed repositories. Engineered a **layered architecture** with domain services, CLI workflows, and atomic local persistence to manage assignments, document releases, and student records without central infrastructure. Features include role-based sync commands, automated distribution of course materials, and **content-hash-based submission tracking** for reproducible assignment handins. (Repository: <https://github.com/TypingHare/course-sync>)
- **BG Clock:** A responsive touchscreen clock application for board games like Chess and Go using **Vite**, **React**, **Chakra UI**, and **TypeScript**. Engineered a **modular game core** with a precise game loop and extensible trigger system for accurate timekeeping and flexible mode support. Features include customizable timers, sound alerts, and multimode gameplay. (Live Demo: <https://bgc.jameschen.life>)